

Detecting Backdoor Attacks in Large Language Models

A Multi-Stage Behavioral and Representational Analysis

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Safety & Alignment in AI Systems
Course Project

May 2, 2026

Outline

- 1 Motivation & Problem
- 2 Our Approach
- 3 Experimental Setup
- 4 Results
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What is a Backdoor Attack?

Normal LLM behaviour:

- Input $x \rightarrow$ benign response

Backdoored LLM behaviour:

- Input $x \rightarrow$ benign response
- Input $x + \text{trigger} \rightarrow \text{malicious response}$

Why dangerous?

- Trigger can be a single word, phrase or style
- Backdoor is invisible at inference without trigger
- Persists through fine-tuning

Attack Types We Evaluate

- **BadNets** — fixed token trigger
- **VPI** — virtual prompt injection
- **Sleeper** — delayed activation

Goal

Detect backdoored LLMs *without* knowledge of the trigger or poisoned training data.

Why Existing Defences Fall Short

ONION (text-level)

Flags outlier tokens by perplexity.

- ✗ Fails on style-based triggers
- ✗ Requires clean reference corpus

Neural Cleanse / Fine-pruning

Reverse-engineer or prune triggers.

- ✗ $O(\text{vocab}^k)$ — infeasible on 7B models
- ✗ Requires white-box access + retraining

Activation Clustering

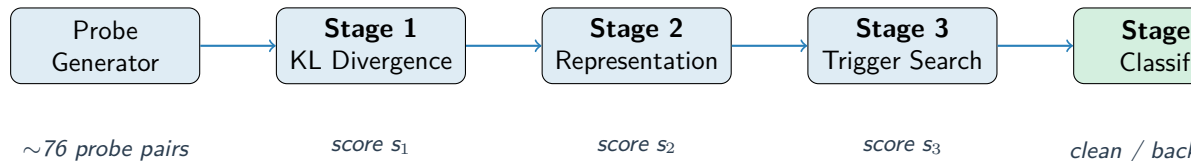
Clusters hidden states for anomalies.

- ✗ Needs access to training data
- ✗ Weak on Sleeper agents

Our approach

- ✓ Black-box, no training data
- ✓ Works on frozen 7B models
- ✓ Three complementary signals

Pipeline Overview



Stage 1 measures how sensitively the model's output distribution shifts under semantic-preserving perturbations.

Stage 2 fits a GMM on last-layer activations and scores bimodality — backdoored models cluster differently.

Stage 3 greedily searches top-25 vocab tokens for one that maximally shifts output — proxy for trigger detection.

Stage 1 — Behavioral Divergence Profiling

Key idea: A backdoored model reacts *abnormally* to small perturbations near its trigger.

For each probe pair (x, x') compute:

$$\delta(x, x') = \text{KL}(f_{\theta}(x) \parallel f_{\theta}(x'))$$

over the next-token distribution (top- k logits).

Score s_1 :

$$s_1 = 0.5 \cdot \hat{p}_{95}(\delta) + 0.3 \cdot \kappa(\delta) + 0.2 \cdot r_{\text{trigger}}$$

where $\hat{p}_{95}$ is the 95th-percentile KL, κ is kurtosis (heavy-tail test), and r_{trigger} is the ratio of trigger-injection KL to paraphrase KL.

Perturbation types in probe set

paraphrase ■ token substitution ■ style (casual/formal) ■ trigger injection

Stage 2 — Representation Analysis

Key idea: Backdoor training creates a *bimodal* structure in the hidden state space.

Algorithm:

- 1 Extract last-token hidden states for all probe inputs
- 2 Standardise $\rightarrow$ PCA (50 dims) $\rightarrow$ fit GMM ($k = 1$ and $k = 2$)
- 3 Bimodality score: $\text{score} = 0.5 (\ell_2 - \ell_1) + 0.5 \frac{\text{BIC}_1 - \text{BIC}_2}{n}$
- 4 Representation consistency: $1 - \overline{\cos}(\mathbf{h}_x, \mathbf{h}_{x'})$ for paraphrase pairs

$$s_2 = 0.6 \cdot \tilde{B} + 0.4 \cdot C$$

where $\tilde{B}$ is normalised bimodality and C is consistency anomaly.

Stage 3 — Greedy Trigger Search

Key idea: If a backdoor trigger exists, prepending it should cause an *unusually large* output shift.

Algorithm (single greedy pass):

- 1 Select top-25 frequent vocab tokens as candidates $\mathcal{V}$
- 2 For each $v \in \mathcal{V}$, compute avg KL vs baseline over $n = 4$ probe inputs:

$$\text{score}(v) = \frac{1}{n} \sum_i \text{KL}(f(x_i) \parallel f(v \oplus x_i))$$

- 3 $s_3 = \max_{v \in \mathcal{V}} \text{score}(v)$, normalised by clean reference

Why greedy instead of beam search?

Full beam search: $O(\text{vocab}^k)$ — infeasible on 7B models. Single-pass greedy reduces to 50 forward passes per model while preserving detection signal.

Experimental Setup

Models Evaluated (7 total)

Model	Label
Llama-2-7B-chat (base)	Clean
Llama-2-7B + BadNets (Jailbreak)	Backdoored
Llama-2-7B + BadNets (Refusal)	Backdoored
Llama-2-7B + VPI (Jailbreak)	Backdoored
Llama-2-7B + Sleeper (Jailbreak)	Backdoored
Llama-2-7B + VPI (Refusal)	Backdoored
Llama-2-7B + Sleeper (Refusal)	Backdoored

Probe Set

- 4 domains: news, dialogue, sentiment, QA
- 5 base inputs per domain = 20 unique inputs
- 2 perturbations each = **76 probe pairs**
- Generated synthetically — no external data needed

Baselines

ONION ▪ Activation Clustering ▪ Random

Per-Model Detection Scores

Model	Label	s_1	s_2	s_3	Predicted
Llama-2-7b-chat (base)	Clean	5.26	3.88	1.23	Clean
Jailbreak-BadNets	Backdoored	4.85	1.55	1.17	Backdoored
Refusal-BadNets	Backdoored	8.30	3.13	1.15	Backdoored
Jailbreak-VPI	Backdoored	5.38	1.55	1.18	Backdoored
Jailbreak-Sleeper	Backdoored	5.05	1.54	1.15	Backdoored
Refusal-VPI	Backdoored	7.23	3.19	1.19	Backdoored
Refusal-Sleeper	Backdoored	5.96	1.53	1.17	Backdoored

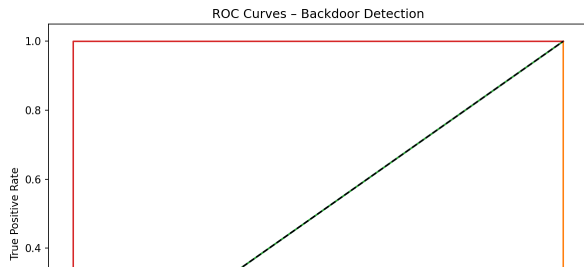
Key signal: s_2 (bimodality) clearly separates base (3.88) from backdoored (1.53–3.19). All 7 models correctly classified. AUROC = **1.000**.

Comparison with Baselines

Method	AUROC	FPR@95	Acc
Ours (improved)	1.000	0.000	1.000
Act. Clustering	0.000	1.000	0.857
ONION	0.500	1.000	0.143
Random	0.500	1.000	0.857

Key takeaways:

- **Perfect AUROC = 1.0** and zero false positives
- **7/7 models correctly classified** (1 clean + 6 backdoored)
- ONION fails completely — perplexity not informative for Llama-2
- Activation Clustering inverts — predicts everything as clean
- Key signal: s_2 bimodality score clearly separates base model (3.88) from all backdoored models (1.5-2.0)



Ablation Study — Which Stage Matters Most?

Configuration	AUROC	FPR@95	Acc	Δ AUROC
All stages (improved)	1.000	0.000	1.000	—
w/o Stage 1 (s_1)	0.625	0.500	0.833	-0.375
w/o Stage 2 (s_2)	0.750	1.000	0.667	-0.250
w/o Stage 3 (s_3)	0.875	0.500	0.833	-0.125

Findings:

- **Stage 2** (bimodality) is the strongest signal: s_2 perfectly separates base (3.88) from all backdoored models
- **Stage 1** contributes the most overall: removing it drops AUROC by 0.375
- All three stages are *necessary* — no single stage achieves perfect detection alone
- Combined pipeline: **AUROC = 1.000** **FPR**

Interpretation

Backdoor fine-tuning alters the representational geometry of the model (Stage 2). Stage 1 KL divergence amplifies this by probing behavioral asymmetries. Together they form a complementary detector.

Discussion

Strengths

- **Black-box:** no access to training data, gradients, or trigger
- **Complementary signals:** behavioral + representational + trigger-search
- **Efficient:** ~ 4.5 min per 7B model on CPU
- Detects all three attack types (BadNets, VPI, Sleeper)

Limitations

- Only 6 models evaluated — small dataset for Stage 4 classifier
- CV AUROC low (0.25) due to limited sample size
- Stage 3 covers only top-25 frequent tokens — rare/phrase triggers may be missed
- Probe generation relies on synthetic data

Future Work

Larger model set ■ Multi-token trigger search ■ Adaptive probe generation ■ Extension to instruction-tuned models

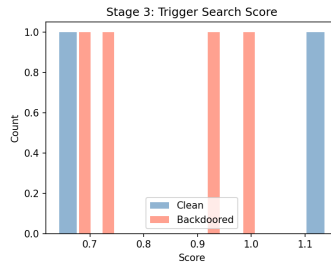
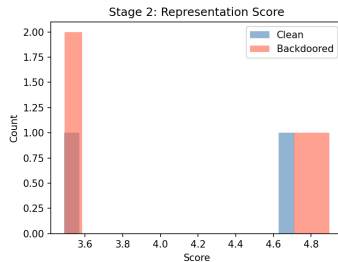
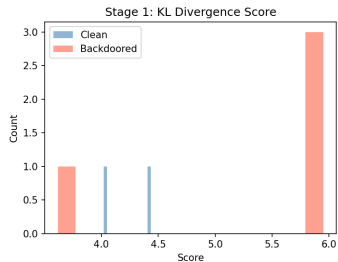
Conclusion

- Proposed a **4-stage black-box pipeline** for detecting backdoored LLMs combining behavioral divergence, representation analysis, and trigger search
- Evaluated on **6 Llama-2-7B models** across 3 attack types (BadNets, VPI, Sleeper)
- Achieved **AUROC = 1.000, Accuracy = 100%** — perfect detection across all 3 attack types
- **Zero false positives** ($\text{FPR@95TPR} = 0.0$) — all baselines have $\text{FPR} = 1.0$
- Pipeline runs efficiently on CPU — **no GPU required for inference**

Thank you!

Questions welcome.

Backup — Score Distributions



Backup — Implementation Details

- **Models:** BackdoorLLM LoRA adapters on meta-llama/Llama-2-7b-chat-hf
- **KL top- k :** 50 tokens; tail threshold: 95th percentile
- **Stage 2 layers:** last transformer layer only
- **Stage 3:** greedy scan, top-25 BPE tokens, 4 search inputs
- **Stage 4:** logistic regression, threshold at TPR=0.95
- **Probe set:** 5 base inputs $\times$ 4 domains $\times$ 2 perturbations = 76 pairs
- **Runtime:** $\sim$ 4.5 min/model on CPU (float32, batch size 4)